

DataLens Pro Version 3.0: An Open-Source AI-Enhanced Streamlit-Based Analytics and Visualization Platform for Automated Business Intelligence

Writer

Sayantana Roy

Department of Computer Science and Engineering

Government College of Engineering and Ceramic Technology (GCECT), Kolkata

Email: sayantanr32@gmail.com

Code Repository

<https://github.com/sayantana/datalens-pro-3.0>

Abstract

Modern organizations generate enormous volumes of structured and semi-structured data daily. Traditional business intelligence platforms such as Tableau and Power BI provide advanced analytical capabilities but often introduce limitations including licensing costs, deployment restrictions, and reduced extensibility. DataLens Pro Version 3.0 is introduced as a fully open-source, AI-enhanced, Streamlit-powered analytics ecosystem designed to democratize advanced data visualization, automated dashboard generation, machine learning insights, and statistical analysis.

The system integrates automated dimensional analysis, dynamic KPI generation, multi-dashboard orchestration, clustering algorithms, forecasting systems, anomaly detection, PCA-driven analytics, and interactive visualization pipelines into a single lightweight platform. Built using Python, Plotly, Streamlit, Scikit-learn, and Statsmodels, the architecture emphasizes modularity, scalability, rapid deployment, and extensibility.

Unlike conventional proprietary analytics tools, DataLens Pro Version 3.0 enables researchers, startups, educators, analysts, and enterprises to construct over 50 automated dashboards and more than 50 intelligent graph visualizations with minimal configuration. The platform additionally supports HTML dashboard export, real-time filtering, automated statistical inference, and machine learning-assisted predictive analytics.

Experimental evaluations demonstrate that the platform significantly reduces dashboard development time while maintaining high analytical flexibility. This research presents the architecture, implementation methodology, algorithmic design, system optimization techniques, machine learning framework, statistical engine, and future research opportunities of DataLens Pro Version 3.0.

Keywords: Streamlit, Business Intelligence, Data Analytics, Machine Learning, Visualization, Open Source, Dashboard Automation, Statistical Computing, Predictive Analytics.

1. Introduction

Business Intelligence (BI) systems have evolved into essential infrastructures for modern organizations. Enterprises increasingly require real-time analytics, automated visualization, predictive forecasting, anomaly detection, and scalable dashboarding systems capable of processing heterogeneous datasets.

Commercial BI tools provide enterprise-grade solutions but often suffer from several limitations:

- High subscription costs
- Vendor lock-in
- Restricted extensibility
- Limited custom AI integration
- Cloud dependency
- Proprietary visualization pipelines

DataLens Pro Version 3.0 addresses these limitations by introducing an open-source analytics ecosystem capable of generating intelligent dashboards automatically using Python-based analytical pipelines.

The platform is implemented using Streamlit, enabling rapid deployment and interactive visualization. Advanced machine learning models and statistical techniques are integrated directly into the dashboarding workflow.

The primary contributions of this work include:

1. Development of an automated dashboard generation framework
2. Integration of AI-driven analytics modules
3. Implementation of scalable visualization pipelines
4. Real-time filtering and dimensional analysis
5. Embedded statistical and machine learning systems
6. HTML report export architecture
7. Open-source extensibility for research and industry

2. Related Work

Several commercial and open-source visualization systems currently dominate the BI ecosystem.

2.1 Tableau

Tableau provides advanced drag-and-drop analytics but operates under a proprietary licensing model. Custom AI integration is comparatively restricted.

2.2 Microsoft Power BI

Power BI integrates strongly with enterprise ecosystems but introduces deployment complexity and cloud-centric dependencies.

2.3 Apache Superset

Apache Superset is open-source and scalable but requires more configuration and backend infrastructure.

2.4 Streamlit Ecosystem

Streamlit simplifies analytical web application development but lacks native enterprise-scale automated analytics frameworks.

DataLens Pro Version 3.0 differentiates itself by combining:

- Streamlit simplicity
 - Automated dashboard intelligence
 - Integrated ML pipelines
 - Statistical inference systems
 - Open-source extensibility
 - Lightweight deployment
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3. System Architecture

The architecture of DataLens Pro Version 3.0 consists of multiple interconnected analytical modules.

3.1 Core Components

The software architecture includes:

- Data ingestion engine
- Auto schema detection
- KPI generator
- Statistical engine
- Visualization engine
- Machine learning module
- Forecasting subsystem
- Clustering subsystem
- Export engine

3.2 Technology Stack

Component	Technology
Frontend	Streamlit

Component	Technology
Visualization	Plotly
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Statistics	SciPy, Statsmodels
Forecasting	ARIMA
Deployment	Python

4. Data Processing Pipeline

The platform automatically processes uploaded CSV and Excel files.

4.1 Automatic Type Detection

The software intelligently categorizes columns into:

- Dimensions
- Measures
- Date columns

The algorithm evaluates:

- Datatype properties
- Uniqueness ratios
- Datetime inference
- Numeric distributions

This significantly reduces manual preprocessing overhead.

4.2 Data Cleaning

The system supports:

- Missing value analysis
 - Null percentage detection
 - Dynamic filtering
 - Aggregation pipelines
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5. Dashboard Generation Engine

One of the most innovative aspects of DataLens Pro Version 3.0 is the automated dashboard generation framework.

5.1 KPI Dashboard

The KPI engine automatically computes:

- Totals
- Means
- Summary metrics
- Performance indicators

5.2 Statistical Dashboards

The statistical module provides:

- Histograms
- Box plots
- Violin plots
- QQ plots
- Correlation heatmaps
- ANOVA testing

5.3 Time Series Analytics

The temporal analytics engine supports:

- Aggregated trend analysis
- Moving summaries
- Forecasting
- ARIMA modeling

5.4 Correlation Systems

The platform dynamically generates correlation matrices using heatmap visualization.

6. Machine Learning Framework

DataLens Pro Version 3.0 integrates multiple machine learning models.

6.1 Linear Regression

The regression system computes predictive relationships between variables using:


$$y = \beta_0 + \beta_1 x + \epsilon$$
 

The platform calculates:

- Predictions
- Residuals

- R^2 scores
- Trend visualizations

6.2 Random Forest Regression

Feature importance is evaluated using ensemble learning methods.

6.3 Gradient Boosting

Gradient boosting regression improves predictive performance through iterative optimization.

6.4 K-Means Clustering

Cluster segmentation is performed using centroid optimization.

$$J = \sum_{i=1}^k \sum_{x \in C_i} ||x - \mu_i||^2$$

6.5 Isolation Forest

The anomaly detection system identifies outliers within multidimensional datasets.

6.6 ARIMA Forecasting

Time-series forecasting is implemented using ARIMA models.

$$\text{ARIMA}(p,d,q)$$

7. Visualization Framework

The visualization engine automatically generates over 50 graph types.

Supported charts include:

- Bar charts
- Line charts
- Scatter plots
- Box plots
- Violin plots
- Heatmaps
- Treemaps
- Area charts
- PCA visualizations
- 3D cluster plots

The system uses Plotly for high-performance interactive rendering.

8. HTML Export System

The software supports dynamic HTML dashboard export.

Exported reports include:

- Data previews
- KPI summaries
- Plotly visualizations
- Correlation heatmaps
- Styled tables

This feature improves portability and reporting capabilities.

9. Performance Optimization

Several optimizations are implemented:

- Cached data loading
- Lazy rendering
- Sampling-based visualization
- Conditional ML execution
- Selective statistical computation

These optimizations significantly improve responsiveness for large datasets.

10. Experimental Evaluation

10.1 Dataset Types

The system was evaluated using:

- Financial datasets
- Sales analytics
- Healthcare datasets
- Research datasets
- Temporal datasets

10.2 Observations

Experimental results indicate:

- Rapid dashboard generation
- Efficient memory utilization
- Accurate ML predictions
- Scalable visualization rendering

11. Applications

DataLens Pro Version 3.0 supports diverse domains.

11.1 Healthcare

- Patient analytics
- Disease forecasting
- Clinical visualization

11.2 Finance

- Market analysis
- Fraud detection
- Revenue forecasting

11.3 Education

- Student performance analytics
- Research visualization

11.4 Industrial Analytics

- Manufacturing monitoring
- Sensor analysis
- Operational dashboards

12. Comparative Analysis

Feature	Tableau	Power BI	DataLens Pro 3.0
Open Source	No	No	Yes
AI Integration	Limited	Moderate	Extensive
Streamlit-Based	No	No	Yes
ML Pipelines	External	Partial	Integrated
Forecasting	Limited	Moderate	Advanced
Cost	High	Medium	Free

13. Future Work

Future versions may include:

- Deep learning modules
 - LLM-powered analytics
 - Real-time streaming dashboards
 - Distributed computation
 - GPU acceleration
 - Natural language querying
 - Autonomous dashboard generation
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14. Conclusion

DataLens Pro Version 3.0 represents a significant advancement in open-source business intelligence systems. By integrating machine learning, statistical analytics, automated visualization, forecasting, clustering, and interactive dashboard generation into a unified Streamlit-based architecture, the platform demonstrates the feasibility of lightweight yet enterprise-capable analytics ecosystems.

The system provides a scalable, extensible, and research-friendly alternative to proprietary BI platforms while remaining accessible to researchers, startups, educators, and enterprises.

The integration of AI-driven analytics pipelines and automated dashboard orchestration establishes DataLens Pro Version 3.0 as a powerful foundation for future intelligent business intelligence research.

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